Analytical Cumulative Examination
February 11, 2005

1. (50) A laser pointer produces 5 mW at 630 nm with a 1 nm FWHM in a beam 1 mm^2.
   a. Calculate the number of photons s^{-1} mm^{-2} nm^{-1} produced by the laser pointer.
   b. A tungsten filament can be <u>approximated</u> as a perfect black body radiator. Calculate the temperature of a tungsten filament needed to produce a maximum wavelength of emission at 630 nm.
   c. The melting point of tungsten is 3410 °C. Determine the spectral radiance at 630 nm (B_λ) for a tungsten filament at its melting point (color temperature = 3410 °C).
   d. Convert B_λ to units of photons s^{-1} sr^{-1} mm^{-2} nm^{-1} and compare this answer to that in a.

$$B_\lambda = \frac{2hc^2}{\lambda^5}\frac{1}{e^{hc/\lambda kT}-1} \qquad \lambda_{\max} = \frac{hc/k}{4.9651T}$$

Use order-of-magnitude values for the physical constants, h, c and k.

2. (50) Using the data below, calculate the following parameters.

Standard Conc (M)	Analyte Response	Blank Response	Std Dev. Blank
1×10^{-5}	1.90	0.4	0.07
5×10^{-6}	1.10	0.4	0.06
2×10^{-6}	0.75	0.5	0.05
1×10^{-6}	0.43	0.3	0.05

   a. Sensitivity.
   b. Signal to noise ratio (SNR) for lowest concentration standard.
   c. Limit of detection (in M) at a SNR =2.
   d. The number of samples was increased by a factor of 10 and averaged together. By what factor would the SNR improve.